

The effect of gauge length on axially incident P-waves measured using fibre optic distributed vibration sensing

Timothy Dean^{1*}, Theo Cuny² and Arthur H. Hartog²

¹Department of Exploration Geophysics, Curtin University, GPO Box U 1987, Perth, WA 6845, Australia, and ²Schlumberger Fibre Optics Technology Centre, Unit 7 The Quadrangle, Abbey Park Industrial Estate, Premier Way, Romsey, SO51 9DL, UK

Received July 2015, revision accepted February 2016

ABSTRACT

Distributed vibration sensing, also known as distributed acoustic sensing, is a relatively new method for recording vertical seismic profile data using a fibre optic cable as the sensor. The signal obtained from such systems is a distributed measurement over a length of fibre referred to as the gauge length. In this paper, we show that gauge length selection is one of the most important acquisition parameters for a distributed vibration sensing survey. If the gauge length is too small, then the signal-to-noise ratio will be poor. If the gauge length is too large, resolution will be reduced and the shape of the wavelet will be distorted. The optimum gauge length, as derived here, is a function of the velocity and frequencies of the seismic waves being measured. If these attributes vary considerably over the depth of a survey, then the use of different gauge lengths is recommended. The significant increases in data quality resulting from the use of multiple gauge length values are demonstrated using field data.

Key words: Fibre optic, distributed acoustic sensing, distributed vibration sensing, vertical seismic profile.

INTRODUCTION

Distributed vibration sensing (DVS), also known as distributed acoustic sensing (DAS), is a relatively new method for recording vertical seismic profile (VSP) data using a fibre optic cable as the sensor, rather than a series of discrete sensors deployed as a borehole seismic array. There are several implementations of the technology (see Hartog, Kotov, and Liokumovich (2013) for a summary), but they are all based on optical time-domain reflectometry (OTDR). An OTDR system operates by launching a pulse of light down a fibre optic cable. As the pulse travels down the cable, small fluctuations of the refractive index within the fibre, frozen-in during the manufacturing process, cause some of the light to be scattered. The small fraction of this light that is scattered within the acceptance angle of the fibre and in the opposite direction to the laser travels back up the fibre where it can be detected by a receiver. The position of the section of the fibre that con-

tains the scatterer can be easily calculated from the recording time, speed of light, and the group refractive index of the glass. The density of the scatterers is far too high for them to be individually distinguished; hence, we extract the average phase of the reflected signal within each section of the fibre occupied by the pulse (Hartog *et al.* 2014) instead.

When the fibre is at rest (blue line in Fig. 1a), the backscattered signal from multiple pulses, although random, is constant (blue line in Fig. 1b). However, when the cable is deformed (red line in Fig. 1a), the travel time to sections of the fibre beyond the point of deformation changes, resulting in a change in the backscattered signal (red line in Fig. 1b). The change in the phase of the backscattered signal has a highly nonlinear transfer function unsuited for seismic applications; thus, the phase difference between two points is used instead (Hartog *et al.* 2014), as shown in Fig. 1c. The physical distance between the two points is referred to as the *gauge length*.

Although we often think of the DVS measurement being the difference between two discrete phase measurements, the reality is that the laser pulse transmitted has an approximately Gaussian amplitude envelope, and thus, the measurement is

*E-mail: tim.dean.geo@gmail.com

Correction added on 4 Sept 2016, after first online publication: Author's affiliation was omitted and is corrected in this version

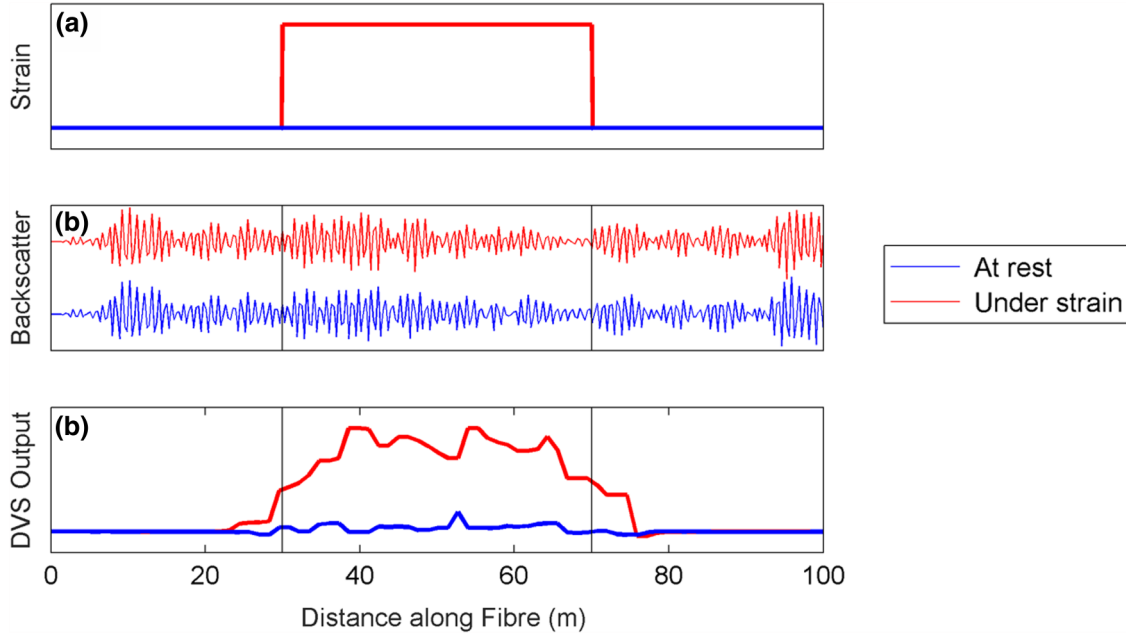


Figure 1 Two synthetic (a) strain, (b) backscatter, and (c) DVS output profiles for the fibre at rest (blue lines) and undergoing localized strain (red lines). The profiles were generated, assuming the acquisition method of Hartog and Kader (2012), which is also detailed in Hartog *et al.* (2014).

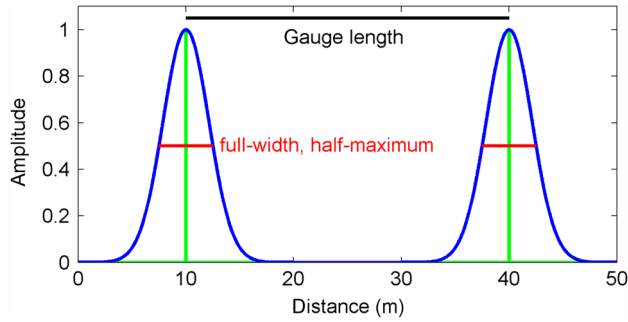


Figure 2 Diagram of the relationship between the gauge length and the pulsewidth (full width at half maximum).

actually the difference between the amplitudes resulting from the application of two Gaussian windows, as shown in Fig. 2. The wavenumber response of the Gaussian window R_p is

$$R_p(k) = e^{-\frac{k^2}{2\sigma_k^2}}, \quad (1)$$

where k is the wavenumber and σ_k is the standard deviation, which is related to the full width at half maximum h by

$$\sigma_k = \frac{2.3548}{2\pi h}. \quad (2)$$

The application of the gauge length GL also has a wavenumber response R_{gl} , which is given by (Dean, Papp, and Hartog 2015)

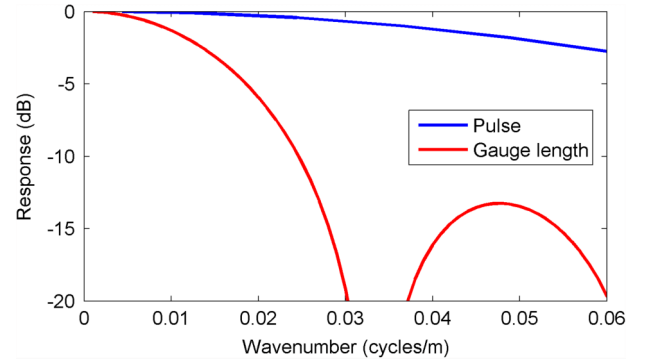


Figure 3 The wavenumber response resulting from a pulsewidth of 5 m and a gauge length of 30 m.

$$R_{gl}(k) = \frac{\sin(\pi k GL)}{\pi k}. \quad (3)$$

Figure 3 shows the wavenumber responses resulting from the application of a pulsewidth of 5 m and a gauge length of 30 m. Clearly, the effect of the gauge length predominates, and numerical experiments show that this is the case unless the pulsewidth is greater than ~ 0.56 times the gauge length. Our experience shows that we are always well above this limit, and thus, in this work, the effect of pulsewidth is neglected, although it will be addressed more comprehensively in future work.

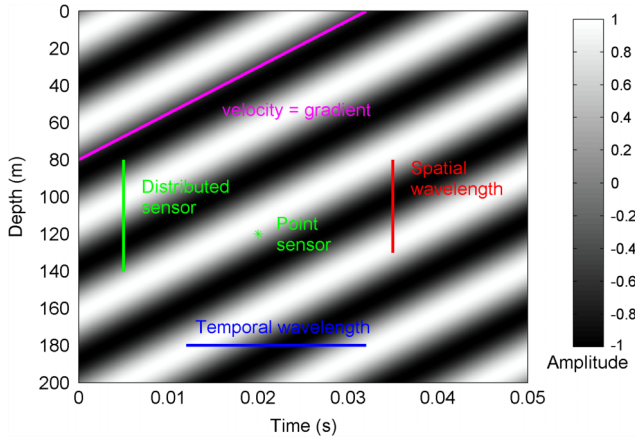


Figure 4 Diagram showing the in-line strain amplitude measured at different positions and times for an axially incident P-wave (velocity = 2500 m/s; frequency = 50 Hz).

Measurements can be made using such systems irrespective of the deployment mode employed, for example, permanently installed cables, either on production tubing (Daley *et al.* 2013; Mateeva *et al.* 2013; Bakku, Wills, and Fehler 2014a; Bakku *et al.* 2014b; Didraga 2015; Longton *et al.* 2015) or cemented behind casing (Mestayer *et al.* 2012; Daley *et al.* 2013; Didraga 2015; Götz *et al.* 2015; Zwartjes and Mateeva 2015), or temporary cables such as fibres embodied within a wireline cable (Frignet and Hartog 2014; Hartog *et al.* 2014). Generally speaking, the better the coupling between the cable and the formation, the better the data quality.

The measurement made using a DVS system is fundamentally different from a conventional point sensor. Figure 4 shows the strain resulting from the passage of a mono-frequency P-wave axially incident to a distributed sensor (i.e., a piece of fibre) at different times and positions. Clearly, a point sensor accurately measures the amplitude of the wave irrespective of its velocity. The distributed sensor, however, is outputting a sum of all the strain amplitudes measured along its length (Dean *et al.* 2015). Thus the output of the sensor will be affected by the spatial wavelength and angle of incidence (Dean *et al.* 2015).

Due to the need to avoid non-linearities in the transfer function, we cannot reduce the gauge length to the extent at which it would effectively become a point sensor; it is thus necessary to employ an appreciable gauge length. The objective of this paper is to determine the effects of the choice of gauge length. We begin by looking at the relationship between gauge

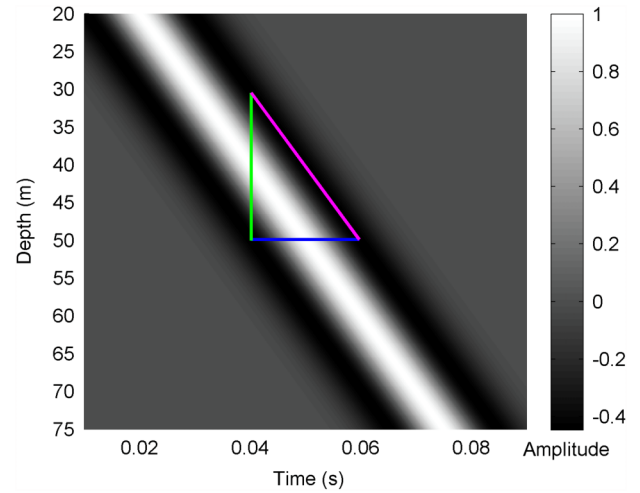


Figure 5 Plot of the strain amplitude resulting from the passage of a 40-Hz Ricker wavelet travelling in the in-line direction with a velocity of 1000 m/s. The blue line is the temporal wavelength, the green line is the spatial wavelength, and the gradient of the magenta line is the velocity.

length and signal-to-noise ratio (SNR) before moving to its relationship with resolution. We then establish an expression to determine the optimum gauge length before demonstrating the effect that the employment of optimized values can have on the resulting data quality.

EFFECTS OF GAUGE LENGTH ON SNR

To determine the effect of gauge length on SNR, we will assume that the signal we are measuring is a P-wave whose direction of arrival is parallel to the sensing fibre (i.e., consistent with a zero-offset VSP survey). The strain amplitude ε of a Ricker wavelet is defined in the time domain as (Ricker 1940, 1944, 1953)

$$\varepsilon(t) = (1 - 2\pi^2 f_p^2 t^2) e^{-\pi^2 f_p^2 t^2}, \quad (4)$$

where f_p is the peak frequency and t is the time. The period or temporal wavelength λ_t (the distance between minima) is

$$\lambda_t = (\sqrt{6}/\pi) / f_p. \quad (5)$$

If we include propagation in the x -direction at velocity v , then equation (4) becomes

$$\varepsilon(t, x) = (1 - 2\pi^2 f_p^2 (t - x/v)^2) e^{-\pi^2 f_p^2 (t - x/v)^2}. \quad (6)$$

A t - x plot of such a wave for arbitrary values of frequency and velocity is shown in Fig. 5; the temporal wavelength is

indicated by the blue line. The gradient of the magenta line is the velocity ($19.5/0.019 = 1000$ m/s), and the spatial wavelength λ_s indicated by the green line is therefore

$$\lambda_s = v\lambda_t. \quad (7)$$

The total change in length of a fibre with gauge length GL at time t will be the integral of equation (6) from $x = \text{start of fibre}$ to $x = \text{end of fibre}$. The maximum strain measured will occur when the centre of the wavelet coincides with the centre of the fibre if, as will be shown later, the gauge length is less than the spatial wavelength. We also need to replace the frequency with the wave number k , which can be calculated by combining equations (5) and (7)

$$k = (\sqrt{6}/\pi) / v\lambda_t. \quad (8)$$

Thus equation (4) becomes

$$\varepsilon(x) = (1 - 2\pi^2 k^2 x^2) e^{-\pi^2 k^2 x^2}. \quad (9)$$

Any strain incident on the fibre causes it to expand or contract by a value equal to the integral of the strain along its length. This change in length, in turn, causes a change in the phase of the backscattered light whose value is proportional to the change in length. The total change in length measured by the fibre ΔL at position x is

$$\Delta L = \int_{-GL/2}^{GL/2} (1 - 2\pi^2 k^2 x^2) e^{-\pi^2 k^2 x^2} dx, \quad (10)$$

which is simply (consistent with the fact that a Ricker wavelet is the second derivative of a Gaussian)

$$\Delta L = x e^{-\pi^2 k^2 x^2} \Big|_{-GL/2}^{GL/2}. \quad (11)$$

An example result for parameters consistent with Fig. 5 is shown in Fig. 6. As the gauge length gets smaller, the value of strain as measured by the fibre ($\varepsilon_f = \Delta L/GL$) approaches the maximum amplitude of the actual strain (i.e., as $GL \rightarrow 0$, $\varepsilon_f \rightarrow \varepsilon_{\max}$, Fig. 6c). The maximum value of ΔL occurs when the gauge length is equal to the distance between the zero crossings on either side of the peak ($\lambda_s/\sqrt{3}$ from the definition of a Ricker wavelet; Fig. 6b) and has the value

$$\Delta L_{\max} = GL e^{-\pi^2 k^2 (GL/2)^2} \quad \text{when } GL = \lambda_s/\sqrt{3}. \quad (12)$$

It is important to note that this relationship will hold for all values of spatial wavelength.

Figure 6 implies that the maximum strain will be measured when the gauge length is as small as possible, but this

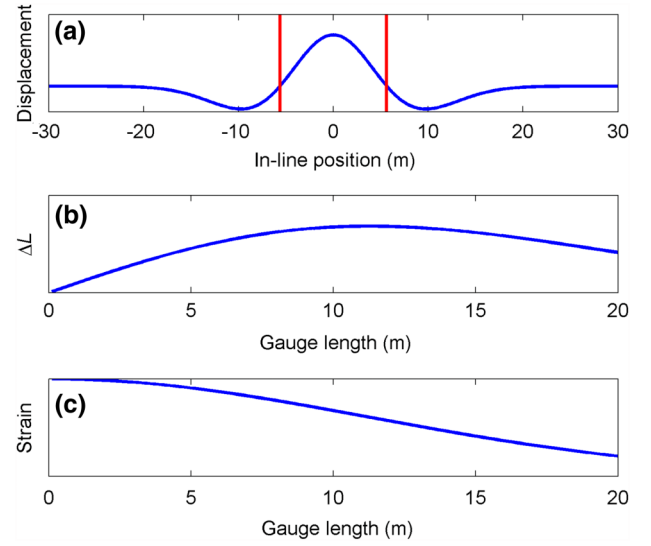


Figure 6 An example Ricker wavelet with the same parameters as those in Fig. 5. The red lines on the strain plot indicate the gauge length that results in the maximum value of ΔL .

ignores the fact that what we are really interested in is the SNR, and thus, the effect of gauge length on noise needs to be included. The noise associated with the strain calculation is a result of the error that arises from the difference between the phases of the backscatter taken from the two sections of the fibre over which the gauge length is being calculated. The error associated with the phase estimate does not change appreciably over the length of the fibre (Dean, Cuny, and Hartog 2015), and thus, the phase error, as well as the error in the value of ΔL that is derived directly from it, is unaffected by gauge length ($E(\Delta L) \propto GL$). The error in the strain measurement ($E(\varepsilon_f)$) therefore will be equal to

$$E(\varepsilon_f) = \frac{E(\Delta L)}{GL}. \quad (13)$$

Note that, unlike ΔL , the error in the fibre-measured strain is affected by the GL ($E(\varepsilon_f) \propto GL$). The SNR of the data will thus be

$$\text{SNR} = \frac{\Delta L/GL}{E(\Delta L)/GL} = \frac{\Delta L}{E(\Delta L)}, \quad (14)$$

and the peak SNR value will occur where the value of ΔL is maximized (equation (12)), i.e., when gauge length/spatial wavelength = $1/\sqrt{3}$ (Fig. 7).

This is not the end of the story, however, as equation (10) is in effect a moving average filter, and moving average filters

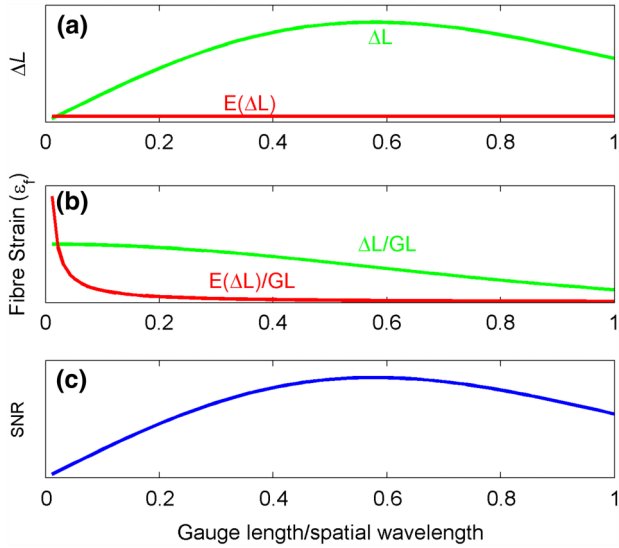


Figure 7 Example plots showing (a) ΔL in green with its associated error in red; (b) fibre-measured strain, colouring as for a; and (c) the resulting SNR for different ratios of gauge length to spatial wavelength.

have a response; it is the effect of this response on resolution that we will now address.

EFFECTS OF GAUGE LENGTH ON RESOLUTION

The frequency response of a moving average filter with length M is known to be

$$\frac{\sin(\pi f_n M)}{M \sin(\pi f_n)}, \quad (15)$$

where f_n is the frequency divided by the sample frequency f_{samp} . For our application, the length is being defined spatially so the filter length must be converted to a temporal length using the velocity. The notches in the response occur at

$$n f_{\text{samp}} / M \text{ where } n \in \mathbb{Z}_+. \quad (16)$$

An example of the responses for four different gauge lengths is shown in Fig. 8. As the gauge length increases, the position of the first notch decreases, intruding into the main bandwidth of the 40-Hz wavelet.

The frequency-domain amplitude of a Ricker wavelet is given by

$$\frac{2}{\sqrt{\pi}} \left(\frac{\omega^2}{\omega_p^3} \right) e^{-\omega^2/\omega_p^2}, \quad (17)$$

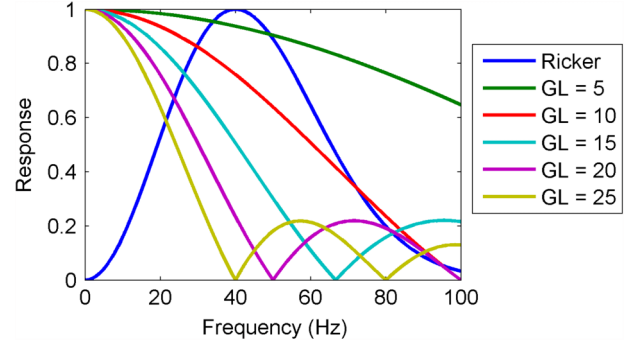


Figure 8 Examples of the frequency response expected from the application of different gauge lengths (in metres) to a Ricker wavelet with a peak frequency of 40 Hz and a velocity of 1000 m/s. The five values correspond to gauge length/wavelength ratios of 0.26, 0.51, 0.77, 1.03, and 1.28, respectively.

where ω is the angular frequency and ω_p the peak angular frequency (Wang 2015). The frequency response of the application of the gauge length to a Ricker wavelet will thus be the product of equations (15) and (17). The results of applying different gauge lengths to a 40-Hz Ricker wavelet in the time domain are shown in Fig. 9. As the gauge length increases, the maximum value of ΔL decreases as does the resulting wavelength due to the increasing attenuation of high frequencies. The intrusion of the first notch of the moving average filter into the bandwidth of the wavelet results in the wavelet having a flat-topped appearance (magenta line, gauge length = 20 m, in Fig. 9) before developing two peaks (olive green line in Fig. 9) as the gauge length further increases.

Figure 10 shows the maximum ΔL value and the resulting wavelength (i.e., the wavelength after the application of the gauge length) for a Ricker wavelet with a centre frequency of 40 Hz and a velocity of 1000 m/s at different gauge length values. As the gauge length increases, the ΔL value increases until it reaches its peak at the value given by equation (12) and shown by the magenta line on Fig. 10a. The resulting wavelength also increases as the increasing gauge length reduces the high-frequency content of the wavelet (Fig. 10b). Once the gauge length is larger than the spatial wavelength (shown by the red lines in Fig. 10), the resulting ΔL wavelet has the two-peaked nature shown in Fig. 9 (olive green line, gauge length = 25 m). When the gauge length is large enough to encompass the complete wavelet, the value at the centre of the ΔL wavelet is simply the integral of the strain over the wavelet length (in this case, 0). Note that, when the gauge length is larger than the spatial wavelength (ratio > 1),

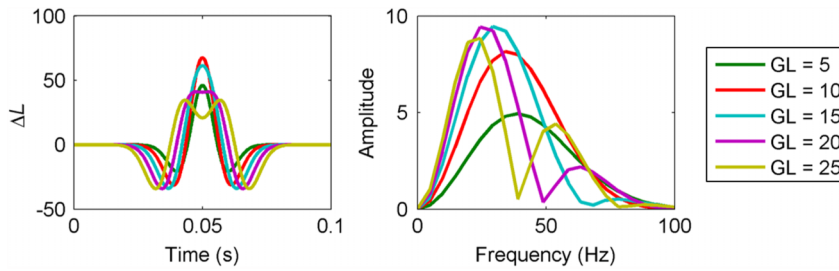


Figure 9 Example wavelets after the application of the gauge length values (in metres) shown in Fig. 8 to a Ricker wavelet with a peak frequency of 40 Hz and a velocity of 1000 m/s.

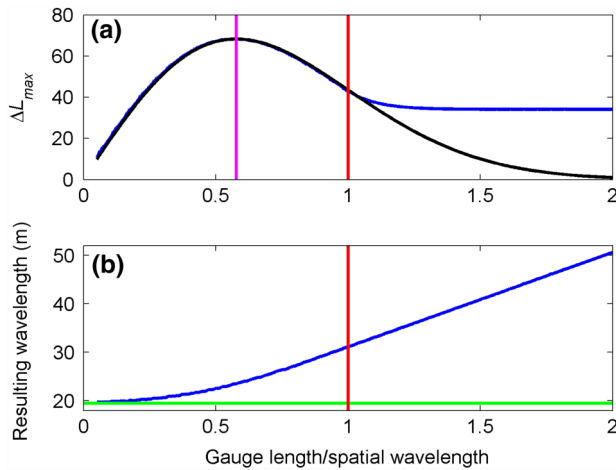


Figure 10 (a) The maximum value of ΔL against the gauge length/spatial wavelength ratio (blue line). The black line is the theoretical value given by equation (9), and the magenta line indicates the position of the maximum value. (b) The resulting measured wavelength (measured between the two minima) against the gauge length/spatial wavelength ratio. The green line is the actual wavelength. The red line on both plots is where gauge length = wavelength.

equation (12) no longer holds and thus the modelled (blue line) and theoretical results (black line) diverge, as the maximum value is no longer located where the centre of the gauge length coincides with the centre of the wavelet.

GAUGE LENGTH OPTIMIZATION

The relationships between SNR and resolution and gauge length can now be combined to determine the optimum gauge length. Figure 11a shows the relationship between SNR and the ratio of gauge length to spatial wavelength. The values within which the SNR is at least 90% of the maximum are shown in the green box; clearly this is the area that we wish to be within. The red box indicates the ratio values where we obtain a double-lobed wavelet and are thus to be avoided.

Figure 11b shows the relationship between the resulting wavelength, as a proxy for resolution, and the ratio of gauge length to spatial wavelength. The green box indicates values where the wavelength is within 3 m (15%) of the actual wavelength. The overlap between these two sections occurs for ratio values between 0.4 and 0.54. The optimum gauge length GL_{opt} is therefore given by

$$GL_{opt} = \frac{\text{ratio } V}{f_p}, \quad (18)$$

where ratio is the desired gauge-length-to-spatial-wavelength ratio. For the example included here, the velocity is 1000 m/s and f_p is 40 Hz, and thus, a desired ratio of 0.4 gives an optimum gauge length of 10 m.

Of course, a Ricker wavelet may not be a particularly good representation of the source signature, particularly if a Vibroseis source is being used. To determine the effect of gauge length on data with a wider bandwidth, we repeated the exercise using a Klauder wavelet (Klauder *et al.* 1960), i.e., the autocorrelation of a Vibroseis sweep, with a frequency range of 8 Hz–80 Hz (i.e., an approximately flat amplitude spectrum between 8 Hz and 80 Hz). The effects of the gauge length, however, will be limited to the recorded data, not to the pilot sweep with which the data are correlated. For the sake of this exercise, we will assume that the pilot has been filtered to be consistent with the data (as it is for conventional systems where the acquisition filters of the recording system are applied to the pilot sweep before correlation) and thus the effect can be modelled using the Klauder wavelet itself. The wider bandwidth results in multiple notches occurring within the wavelet bandwidth for even relatively short gauge lengths (Fig. 12). When we look at the relationship between SNR and gauge-length-to-spatial-wavelength ratio (Fig. 13), we notice some obvious differences from the Ricker wavelet results shown in Fig. 11 but also some similarities. Firstly, the SNR plot is not flat beyond a ratio value of 1; this is because the large high-frequency content of the Klauder wavelet results

Figure 11 The resulting (a) SNR and (b) wavelength for different ratios of gauge length to spatial wavelength. The green box in (a) indicates the range of ratios where $\text{SNR} > 90\%$ of the maximum, and the box in (b) indicates the range of wavelengths within 3 m of the actual wavelength. The red box in (a) indicates areas within which the wavelet shape is distorted.

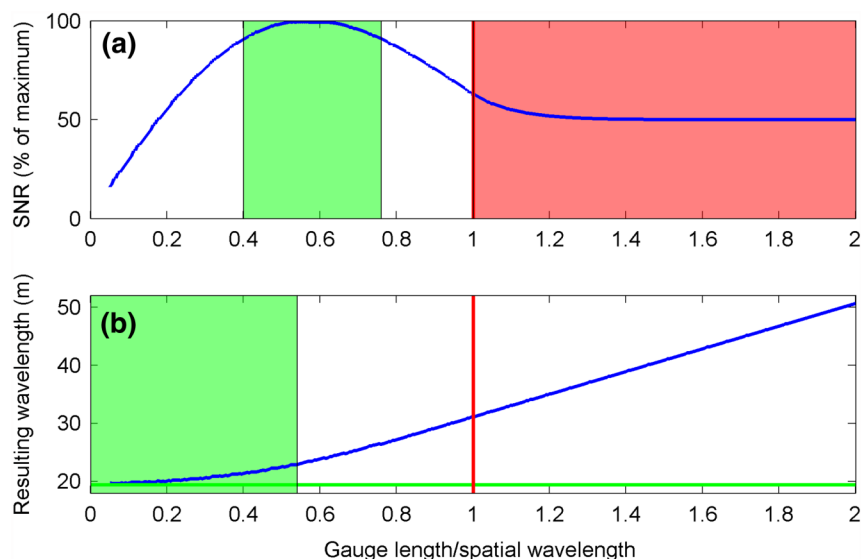
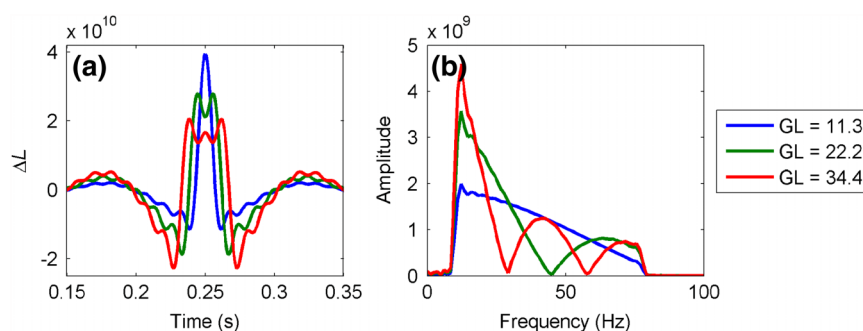


Figure 12 Example wavelets after the application of three different gauge lengths to an 8-Hz to 80-Hz Klauder wavelet with a velocity of 1000 m/s.



in multiple notches being included, further reducing the energy of the wavelet and, thus, the SNR. If we were to increase the range of ratio values shown in Fig. 11, then this effect would also be evident. The SNR plots do, however, have a peak at the same ratio value of ~ 0.6 and have similar optimum gauge length ranges (the ranges shown by the green boxes in Figs. 11a and 13a differing by 0.027) and wavelength (the ranges shown by the green boxes in Figs. 11b and 13b differing by 0.003).

FIELD DATA

The data shown here were acquired as part of a walk-above survey over a deviated well. Two conventional borehole sensors were connected to a hybrid heptacable (Varkey *et al.* 2008) and eight source positions occupied, as shown

in Fig. 14. Data were recorded using the full length of the cable, i.e., from the top of the shallowest sensor to the surface, but for simplicity, only a section of the cable for each source position is shown here (indicated by the colours of the well points shown in Fig. 14). A single 48000-lb vibrator was used as the source sweeping a 12-second, 6-Hz to 96-Hz linear sweep. Five sweeps were acquired at each source position.

The apparent velocities of the data combined from all the shot points ranged from 2900 m/s to 5900 m/s, a factor of nearly two; hence, a single gauge length value is unlikely to be optimum across all depths (the default gauge length used during acquisition was 13.6 m). There are three different stages during DVS data acquisition where the gauge length can be applied, namely, within the optics, within the electronics, or within processing. Our system utilizes the latter approach,

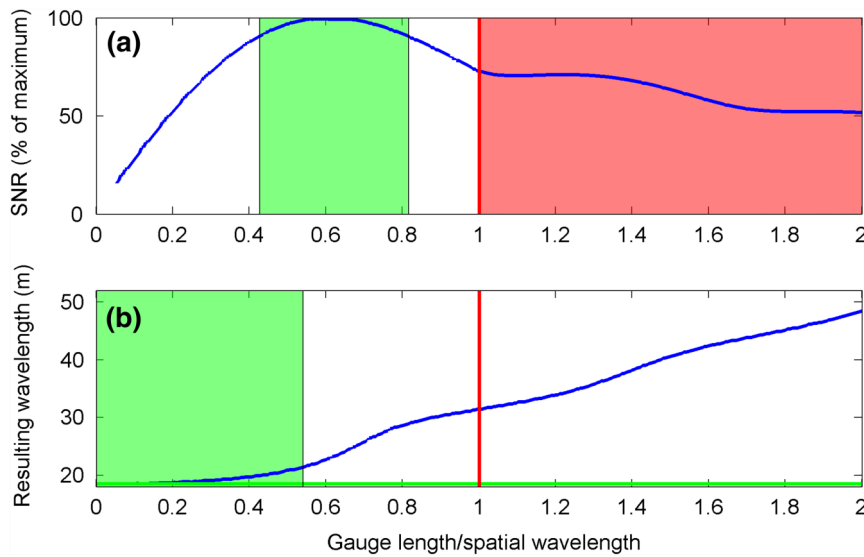


Figure 13 The resulting (a) SNR and (b) wavelength for different ratios of gauge length to spatial wavelength for an 8-Hz to 80-Hz Klauder wavelet with a velocity of 1000 m/s. The green box in (a) indicates the range of ratios where SNR > 90% of the maximum, and the box in (b) indicates the range of wavelengths within 3 m of the actual wavelength. The red box in (a) indicates areas within which the wavelet shape is distorted.

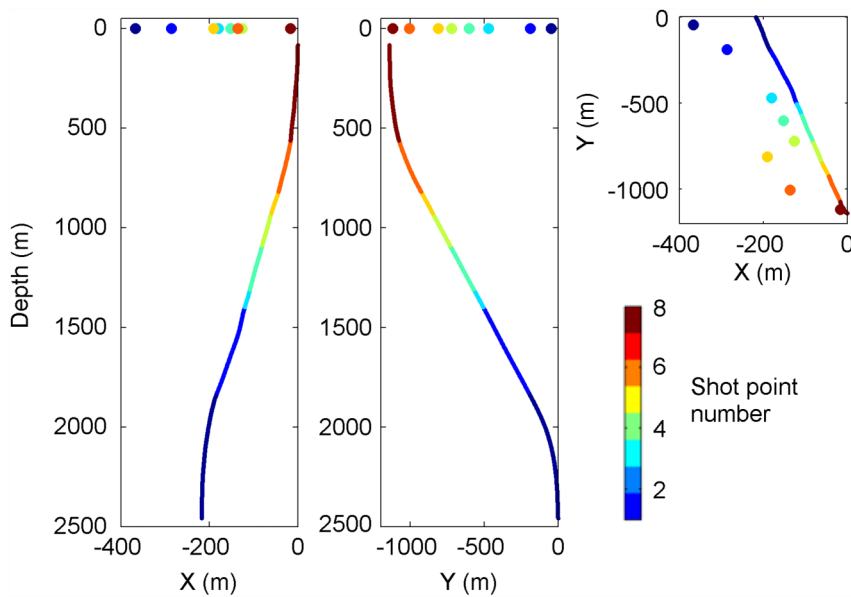


Figure 14 Geometry of the walkabove survey. The colours correspond to the different shot-point numbers.

and by recording raw phase data, we can apply different gauge lengths after acquisition. Figure 15 shows the effect on the data from the shot point with the greatest offset when the data were reprocessed with the optimum gauge length of 38 m (calculated using equation (18) based on a gauge-length-to-wavelength ratio value of 0.5). Before reprocessing (Fig. 15a), the SNR was poor, averaging around 3.5. Reprocessing the data with the optimum gauge length (Fig. 15b) more than

doubled the SNR and resulted in the reflections in the data becoming visible.

Figure 16 shows the full dataset before and after reprocessing with one of three optimum gauge lengths. The resulting improvement in quality is clear; when processed with the default gauge length, there is noise evident above the first breaks. After reprocessing, the noise level is much lower and multiple reflections have become visible.

Figure 15 Data from the furthest shot point flattened on the first break times for gauge lengths of (a) 13.6 m (default) and (b) 38 m (optimized value). The apparent velocity in this section of the well was ~ 4600 m/s. The lower plots show the SNR calculated as the peak amplitude value at the first break divided by the RMS of the noise in a time window at the beginning of the record.

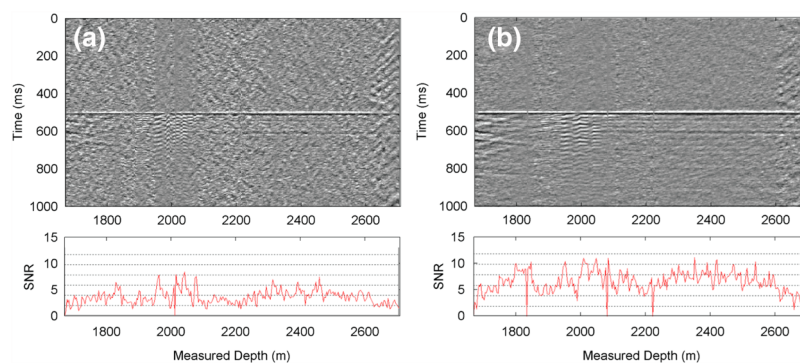
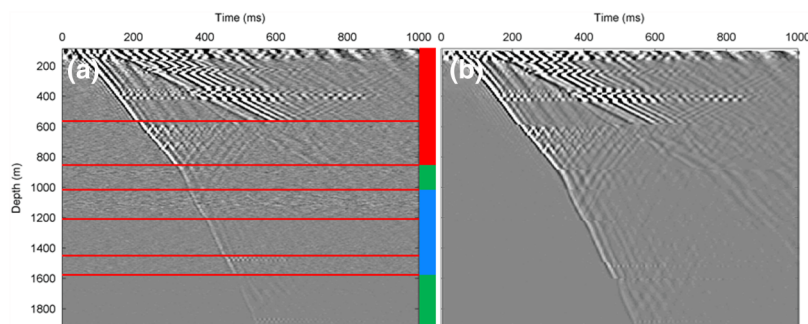


Figure 16 Data (a) before and (b) after re-processing with three different optimum gauge length values calculated on the basis of a ratio value of 0.5. The different shot points are indicated by the red lines on (a). The coloured bars between the plots indicate the gauge lengths (red is 24 m, blue is 38 m, and green 54 m) applied to the data shown in (b).



DISCUSSION AND CONCLUSION

The work presented here clearly shows that gauge length selection is one of, if not the, most important acquisition parameters for a DVS survey. If the gauge length is too small, then the SNR will be poor. If the gauge length is too large, although SNR will likely be relatively high (but not maximized), resolution will be reduced. Finally, if the limit given by equation (18) for a ratio of gauge length to wavelength of 1 is exceeded, then the shape of the wavelet will be distorted.

When the data acquired have a range of apparent velocities, either due to the survey configuration and/or the geology, then a range of appropriately selected gauge lengths should be applied to maximize SNR and resolution over the full length of the well. If the DVS system is being used to determine a time–depth relationship, an application for which it is well suited, then the use of different gauge lengths is important as it will help stabilize the first-break wavelets.

The number of gauge length values given in the literature is limited. Mateeva *et al.* (2014) state that the typical range is between 7 m and 35 m, and although presented results were acquired using a gauge length of 7 m, acknowledge that this length is likely to result in reduced SNR. At the extremes of this range are 10 m, used by Hartog *et al.* (2014), and 40 m, used by Bakku *et al.* (2014a,b). Our experience to date and

the results presented here suggest that values between 20 m and 60 m are likely to be optimum. Theoretically speaking, if the gauge and pulse lengths could be reduced, then the DVS measurement would become, in essence, a strain measurement, but reducing the gauge length to less than 8 m results in the relationship between phase and strain becoming non-linear as the gauge length approaches the laser pulse length. Thankfully, typical seismic wavelengths are large enough that small gauge lengths are unlikely to be required.

ACKNOWLEDGEMENTS

The authors would like to thank their colleagues at the Schlumberger Fibre Optics Technology Centre, Schlumberger Gould Research Centre, and Schlumberger Wireline Operations; Lundin Petroleum for allowing us to show their data; and the editor and three anonymous reviewers whose comments helped strengthen this paper.

REFERENCES

- Bakku S.K., Wills P. and Fehler M. 2014a. Monitoring hydraulic fracturing using distributed acoustic sensing in a treatment well. 84th SEG annual meeting, Denver, USA, Extended Abstracts, 5003–5008.

- Bakku S.K., Wills P., Fehler M., Mestayer J., Mateeva A. and Lopez J. 2014b. Vertical seismic profiling using distributed acoustic sensing in a hydrofrac treatment well. 84th SEG annual meeting, Denver, USA, Expanded Abstracts, 5024–5028.
- Daley T.M., Freifeld B.M., Ajo-Franklin J., Dou S., Pevzner R., Shulakova V. *et al.* 2013. Field testing of fiber-optic distributed acoustic sensing (DAS) for subsurface seismic monitoring. *The Leading Edge* 32, 699–706.
- Dean T., Cuny T. and Hartog A. 2015. Determination of the optimum gauge length for borehole seismic surveys acquired using distributed vibration sensing. 77th EAGE meeting, Madrid, Spain, Expanded Abstracts, Tu N118 16.
- Dean T., Hartog A., Papp B. and Frignet B. 2015. Fibre optic based vibration sensing: Nature of the measurement. In: *3rd EAGE Workshop on Borehole Geophysics*, Athens, Greece, Expanded Abstracts, BG04.
- Dean T., Papp B. and Hartog A. 2015. Wavenumber response of data recorded using distributed fibre-optic systems. In: *3rd EAGE Workshop on Borehole Geophysics*, Athens, Greece, Expanded Abstracts, BGP07.
- Didraga C. 2015. DAS VSP recorded simultaneously in cemented and tubing installed fiber optic cables. In: *77th EAGE Conference and Exhibition*, Madrid, Spain, Expanded Abstracts, Tu N118 14.
- Frignet B.G. and Hartog A.H. 2014. Optical vertical seismic profile on wireline cable. *SPWLA 55th Annual Logging Symposium*, Abu Dhabi, United Arab Emirates, 91575N.
- Götz J., Lüth S., Henniges J. and Reinsch T. 2015. Using a fibre optic cable as distributed acoustic sensor for vertical seismic profiling at the Ketzin CO₂ storage site. *77th EAGE Conference and Exhibition*, Madrid, Spain, Expanded Abstracts, Th P2 13.
- Hartog A., Frignet B., Mackie D., and Clark M. 2014. Vertical seismic optical profiling on wireline logging cable. *Geophysical Prospecting* 62, 693–701.
- Hartog A. and Kader K. 2012. Distributed fiber optic sensor system with improved linearity. US Patent 20120067118A1.
- Hartog A.H., Kotov O.I. and Liokumovich L.B. 2013. The optics of distributed vibration sensing. *2nd EAGE Workshop on Permanent Reservoir Monitoring*, Stavanger, Norway, Expanded Abstracts, Th–01–04.
- Klauder J.R., Price A.C., Darlington S. and Albersheim W.J. 1960. The theory and design of chirp radars. *Bell System Technical Journal* 39, 745–808.
- Longton M., Strudley A., White D., Daley T.M. and Miller D.E. 2015. Simultaneous acquisition of distributed acoustic seismic surveys with single-mode and multi-mode fiber optic cables at Aquistore CO₂ storage site and a comparison with 3C Geophones. *3rd EAGE Workshop on Permanent Reservoir Monitoring*, Oslo, Norway, Expanded Abstracts, Th A04.
- Mateeva A., Lopez J., Potters H., Mestayer J., Cox B., Kiyashchenko D. *et al.* 2014. Distributed acoustic sensing for reservoir monitoring with vertical seismic profiling. *Geophysical Prospecting* 62, 679–692.
- Mateeva A., Mestayer J., Yang Z., Lopez J., Wills P., Roy J. *et al.* 2013. Dual-well 3D VSP in deepwater made possible by DAS. 83rd SEG annual meeting, Houston, USA, Expanded Abstracts, 5062–5066.
- Mestayer J., Grandi Karam S., Cox B., Wills P., Mateeva A., Lopez J. *et al.* 2012. Distributed acoustic sensing for geophysical monitoring. *74th EAGE Annual Conference and Exhibition*, Copenhagen, Denmark, Y005.
- Ricker N. 1940. The form and nature of seismic waves and the structure of seismograms. *Geophysics* 5, 348–366.
- Ricker N. 1944. Wavelet functions and their polynomials. *Geophysics* 33, 314–323.
- Ricker N. 1953. The form and laws of propagation of seismic wavelets. *Geophysics* 18, 10–40.
- Varkey J., Mydur R., Sait N., Wijnberg W., Kunathikom S. and Darpi M. 2008. Optical fiber cables for wellbore applications. US Patent US7324730 B2.
- Wang Y. 2015. Frequencies of the Ricker wavelet. *Geophysics* 80, A31–A37.
- Zwartjes P. and Mateeva A. 2015. Multi-Fibre DAS walk-away VSP at Kapuni. *3rd EAGE Workshop on Borehole Geophysics*, Athens, Greece, Expanded Abstracts, BG05.